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DATABASE DESIGN DOCUMENT

"AI Crop Disease Detection and Treatment System"

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1. General Information

1.1 Introduction

Database Design is the process of producing a detailed data model of a database. This document describes the logical and physical design of the crop_disease_db database used in the AI Crop Disease Detection and Treatment System. Since no farmer login is required, the database stores uploaded images, AI prediction results, crop information, disease data, and treatment recommendations only.

The design process involves: determining relationships between data elements, dividing information into normalized tables, specifying primary and foreign keys, and applying normalization rules to ensure data integrity and eliminate redundancy.

1.2 Purpose

- Define the complete structure of the crop_disease_db database
- Describe each table, its fields, data types, lengths, constraints, and relationships
- Provide an Entity-Relationship (ER) diagram for visual representation
- Ensure the design supports all functional requirements in the SRS
- Serve as a reference for developers during implementation and testing

1.3 System Overview

The AI Crop Disease Detection and Treatment System is a web-based application. No login is required for farmers — any user can upload a crop leaf image and instantly receive disease detection results and treatment recommendations. A separate secured Admin interface allows management of crop, disease, and treatment records. The database is implemented in MySQL or SQLite, accessed via Python Flask and SQLAlchemy ORM.

1.4 Acronyms and Abbreviations

Acronym	Full Form
DB	Database
PK	Primary Key

Acronym	Full Form
FK	Foreign Key
ER	Entity-Relationship
DDD	Database Design Document
SRS	Software Requirements Specification
CNN	Convolutional Neural Network
AI	Artificial Intelligence
DBMS	Database Management System
NULL	No value / Optional field
NOT NULL	Mandatory field
DDL	Data Definition Language

2. Database Identification and Description

2.1 Database Identification

Database Name: crop_disease_db

Database Type: Relational Database (MySQL / SQLite)

Version: 1.0

Note: No User/Farmer table is required since the system does not need farmer login or registration.

2.2 Systems Using the Database

System / Module	Database Operations Performed
Farmer (User) Module	INSERT into Image; SELECT from Prediction for displaying results
Admin Module	INSERT / UPDATE / DELETE in Crop, Disease, and Treatment tables
Image Processing Module	INSERT into Image table; UPDATE image status

System / Module	Database Operations Performed
AI Disease Detection Module	SELECT from Disease; INSERT into Prediction table
Recommendation Module	SELECT from Disease and Treatment tables

2.3 Database Table Structure

The database consists of 5 tables. Each table is described with field names, descriptions, data types, lengths, constraints, and remarks.

Table 1: Image

Stores information about crop leaf images uploaded by users for disease detection.

Field Name	Field Description	Data Type	Length	Constraints	Remarks
Image_ID	Unique image identifier	INT	-	PK, NOT NULL, AUTO_INCREMENT	Primary Key
Image_Path	Server file path of image	VARCHAR	255	NOT NULL	Path to /uploads/
Upload_Date	Date and time of upload	DATETIME	-	DEFAULT CURRENT_TIMESTAMP	Auto-set on insert
Status	Processing status	VARCHAR	20	DEFAULT 'Pending'	Pending / Processed / Failed

Table 2: Crop

Stores the list of supported crops in the system.

Field Name	Field Description	Data Type	Length	Constraints	Remarks
Crop_ID	Unique crop identifier	INT	-	PK, NOT NULL, AUTO_INCREMENT	Primary Key
Crop_Name	Name of the crop	VARCHAR	100	NOT NULL, UNIQUE	e.g., Tomato, Brinjal
Description	Brief crop description	TEXT	-	NULL	Optional information
Season	Growing season	VARCHAR	50	NULL	e.g., Kharif, Rabi

Table 3: Disease

Stores information about crop diseases detectable by the AI model.

Field Name	Field Description	Data Type	Length	Constraints	Remarks
Disease_ID	Unique disease identifier	INT	-	PK, NOT NULL, AUTO_INCREMENT	Primary Key
Disease_Name	Name of the disease	VARCHAR	150	NOT NULL, UNIQUE	e.g., Tomato

Field Name	Field Description	Data Type	Length	Constraints	Remarks
					Late Blight
Symptoms	Visible symptoms description	TEXT	-	NOT NULL	Displayed to users
Cause	Cause of disease	VARCHAR	200	NULL	Fungal / Bacterial / Viral
Crop_ID	Linked crop	INT	-	FK (Crop.Crop_ID), NOT NULL	References Crop table

Table 4: Treatment

Stores treatment and prevention recommendations for each disease.

Field Name	Field Description	Data Type	Length	Constraints	Remarks
Treatment_ID	Unique treatment identifier	INT	-	PK, NOT NULL, AUTO_INCREMENT	Primary Key
Disease_ID	Linked disease	INT	-	FK (Disease.Disease_ID), NOT NULL	References Disease table
Fertilizer	Recommended fertilizer	VARCHAR	200	NULL	e.g., NPK 20-20-20
Pesticide	Recommended pesticide	VARCHAR	200	NULL	e.g., Mancozeb 75% WP

Field Name	Field Description	Data Type	Length	Constraints	Remarks
Prevention	Prevention tips	TEXT	-	NULL	Cultural / organic measures
Dosage	Application dosage	VARCHAR	100	NULL	e.g., 2g per litre water

Table 5: Prediction

Stores AI disease detection results for each uploaded image.

Field Name	Field Description	Data Type	Length	Constraints	Remarks
Prediction_ID	Unique prediction identifier	INT	-	PK, NOT NULL, AUTO_INCREMENT	Primary Key
Image_ID	Linked uploaded image	INT	-	FK (Image.Image_ID), NOT NULL	References Image table
Disease_ID	Predicted disease	INT	-	FK (Disease.Disease_ID), NULL	NULL if crop is healthy
Accuracy	Model confidence score	FLOAT	-	NOT NULL	Value 0.0 to 1.0
Result_Label	Human-readable result	VARCHAR	150	NOT NULL	e.g., Tomato Late Blight

Field Name	Field Description	Data Type	Length	Constraints	Remarks
Predicted_At	Timestamp of prediction	DATETIME	-	DEFAULT CURRENT_TIMESTAMP	Auto-set on insert

3. ER Diagram

The ER Diagram below illustrates the 5 entities, their attributes, and relationships in the crop_disease_db database. Rectangles represent entities, ovals represent attributes (underlined oval = Primary Key attribute), and diamonds represent relationships. Numbers on connecting lines indicate cardinality (1 = one, N = many).

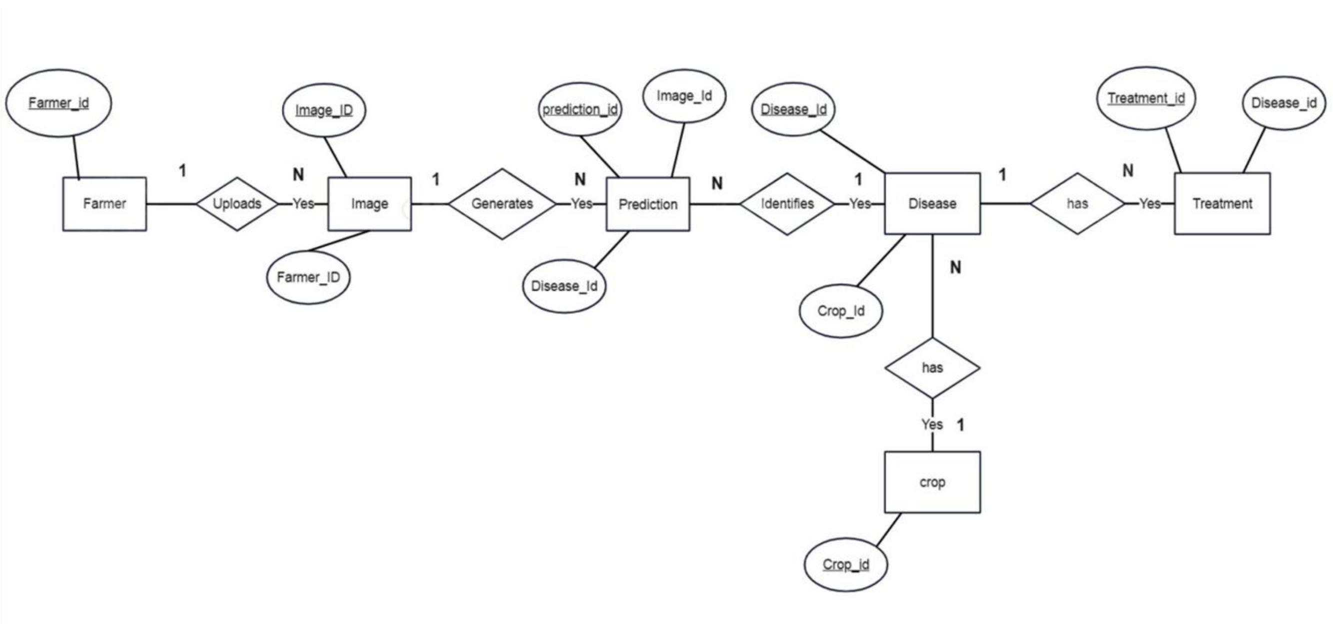

Fig 3.1: ER Diagram — AI Crop Disease Detection and Treatment System

3.1 Entity Descriptions

Entity	Attributes	Description
Image	Image_ID (PK), Image_Path, Upload_Date, Status	Crop leaf image uploaded by any user (no login required)
Prediction	Prediction_ID (PK), Image_ID (FK), Disease_ID (FK), Accuracy, Result_Label, Predicted_At	AI model result linked to an uploaded image
Disease	Disease_ID (PK), Disease_Name, Symptoms, Cause, Crop_ID (FK)	Crop disease detectable by the AI model
Treatment	Treatment_ID (PK), Disease_ID (FK), Fertilizer, Pesticide, Prevention, Dosage	Treatment recommendation for a disease
Crop	Crop_ID (PK), Crop_Name, Description, Season	Type of crop supported by the system

3.2 Relationship Descriptions

Relationship	Entities Involved	Cardinality	Description
has	Image — Prediction	1 : 1	Each uploaded image produces exactly one prediction result
identifies	Prediction — Disease	N : 1	Many predictions may identify the same disease
affects	Disease — Crop	N : 1	Many diseases can affect the same crop type
has	Disease — Treatment	1 : N	One disease can have multiple treatment records

4. Database Normalization

4.1 First Normal Form (1NF)

All tables satisfy 1NF: each table has a primary key, all column values are atomic (no multi-valued attributes in a single field), and each row is unique.

4.2 Second Normal Form (2NF)

All tables satisfy 2NF: all non-key attributes are fully functionally dependent on the entire primary key. Since all primary keys are single-column, no partial dependencies exist.

4.3 Third Normal Form (3NF)

All tables satisfy 3NF: no transitive dependencies exist between non-key attributes. Treatment data is stored separately in the Treatment table and referenced via Disease_ID, rather than being duplicated in the Disease table.

4.4 Anomalies Handled

Anomaly Type	Problem Scenario	How the Design Handles It
Update Anomaly	Disease name stored in multiple rows	Disease stored once in Disease table; referenced by Disease_ID in other tables
Insertion Anomaly	Cannot add treatment without disease record	Treatment has FK to Disease; Disease record must exist before adding treatment
Deletion Anomaly	Deleting disease removes its treatments	Handled via ON DELETE CASCADE or application-level constraint